Sentra — Cron Job Monitoring Demo

Step-by-step commands to create a scheduled job, watch Sentra monitor it, simulate failures, and delete it — with the expected output of every step.

How it works (say this first)

A scheduled (cron) job cannot be observed from outside: nothing on the network tells you whether last night's backup ran. So the job reports to Sentra itself. Each time it finishes successfully, it calls a secret **ping URL**. Sentra does not poll on a timer: it re-checks a job right after each ping, and again exactly when the next ping is due (its deadline). If pings stop, or the job reports a failure, the check turns **DOWN**, an **incident** opens, and alerts are sent. This is called a **dead man's switch**, and it is the **push** model: the monitored job calls Sentra, Sentra never logs into the job's machine.

```
Windows Task Scheduler (every minute)
   └─ runs demo-cron-job.ps1  ── zips the docs folder (a fake "backup")
        ├─ start     ──►  /ping/<token>/start
        ├─ success   ──►  /ping/<token>          ──►  Sentra: "last ping just now" → HEALTHY
        └─ failure   ──►  /ping/<token>/fail     ──►  Sentra: DOWN + incident (immediately)
   no ping at all for (period + allowed delay)   ──►  Sentra: DOWN + incident
```

What the demo shows

1. **Part A** — Start Sentra (database, backend, dashboard)
2. **Part B** — Create the job: Steps 1–6
3. **Part C** — Monitoring and failure demos: Steps 7–8
4. **Part D** — Delete the job: Step 9
5. **Part E** — Troubleshooting

Part A — Start Sentra (do this before the teacher arrives)

Use three separate PowerShell windows. Leave them all open during the demo.

A1. Start the database

Open **Docker Desktop** and wait until it says it is running. Then:

```
cd C:\Users\shubh\OneDrive\Documents\Sentra
docker compose up -d
```

MEANING

Starts PostgreSQL in a container on port 55432. All checks, runs and incidents are stored here.

Expected: `Container sentra-db Started` (or `Running` if it was already up).

A2. Start the backend (window 1)

```
cd C:\Users\shubh\OneDrive\Documents\Sentra\backend
mvn spring-boot:run "-Dspring-boot.run.profiles=dev" `
    "-Dspring-boot.run.jvmArguments=-Xmx512m" `
    "-Dspring-boot.run.arguments=--server.port=8081"
```

The `` ` `` (backtick) at the end of a line means "the command continues on the next line".

MEANING

Runs the Spring Boot server on port 8081. It contains the scheduler that runs every check when it is due.

Expected: after ~15 seconds the log ends with `Tomcat started on port 8081` and `Started SentraApplication`. Leave this window open.

If `mvn` is not recognised, run this first in the same window:

```
$machine = [Environment]::GetEnvironmentVariable("Path","Machine")
$user    = [Environment]::GetEnvironmentVariable("Path","User")
$env:Path = "$machine;$user"
```

A3. Start the dashboard (window 2)

```
cd C:\Users\shubh\OneDrive\Documents\Sentra\frontend
npm run dev
```

Expected: `VITE ... ready` and a line `Local: http://localhost:5173/` (or 5174 if 5173 is busy). Open that address in the browser and log in.

A4. Check the backend is alive (window 3 — use this window for every step below)

```
curl.exe http://localhost:8081/api/v1/ping
```

Expected: `{"status":"ok"}`

Part B — Create the demo job

Step 1. Create the check in Sentra (browser)

1. In the dashboard, click **+ Add check**.
2. Choose the ⏱ **Cron job** type at the top of the form.
3. Name: `Demo backup job`
4. Job runs every: **1 minute** · Allowed delay (minutes): **1**
5. Click **Add check**.

MEANING

Tells Sentra: "expect a ping from this job every minute; if none arrives within 1 minute plus 1 minute of slack, raise the alarm."

Expected: the check's page opens with a **Ping URL** such as `http://localhost:5173/ping/AbC123xyz...`. State is **PENDING**, "Last ping: none yet". Copy the **token** — the long text after `/ping/`.

"Sentra generated a random, unguessable token. Possessing this URL is the job's only credential."

Step 2. Save the token in a variable

```
$token = "PASTE-YOUR-TOKEN-HERE"
```

Expected: no output.

Step 3. Send one ping by hand

```
curl.exe http://localhost:8081/ping/$token
```

MEANING

You pretend to be the job and report "I ran". This proves the URL and token are correct.

Expected: `OK`. In the browser, "Last ping" shows **just now** and within ~5 s the state becomes **HEALTHY**.
`404` means the token was copied wrongly — redo Step 2.

Step 4. Create the scheduled task (runs every minute)

```
$script = "C:\Users\shubh\OneDrive\Documents\Sentra\scripts\demo-cron-job.ps1"
$run  = "conhost.exe --headless powershell.exe -NoProfile -ExecutionPolicy Bypass"
$run += " -File $script -Token $token"
schtasks /Create /TN "SentraDemoJob2" /SC MINUTE /MO 1 /TR $run /F
```

MEANING

Windows Task Scheduler is Windows' version of cron. The task runs the demo script every minute, hidden (no window). The script zips the project's `docs` folder as a pretend backup, then pings Sentra.

Expected: `SUCCESS: The scheduled task "SentraDemoJob2" has successfully been created.`

Step 5. Allow the task to run on battery

```
$settings = New-ScheduledTaskSettingsSet -AllowStartIfOnBatteries -DontStopIfGoingOnBatteries
Set-ScheduledTask -TaskName "SentraDemoJob2" -Settings $settings
```

MEANING

By default Windows skips scheduled tasks when a laptop is unplugged — the job would silently never run.

Expected: a short description of the task, no red error text.

"This is exactly the kind of silent failure Sentra exists to catch: the job simply never ran, and nothing crashed."

Step 6. Run it once now and confirm

```
schtasks /Run /TN "SentraDemoJob2"
Get-Content "$env:TEMP\sentra-demo-backups\job.log" -Tail 3
```

Expected: `SUCCESS: Attempted to run the scheduled task "SentraDemoJob2".`
A new log line like `2026-10-28 18:30:05Z OK docs-20261028-183004.zip`.
In the dashboard the check is **HEALTHY**, and "Last ping" refreshes every minute on its own from now on.

Part C — Monitoring and failure demos

What to point at on the dashboard

On screen	What it means
PENDING	Created, no ping yet — Sentra does not alert for a job that has not had a chance to run.
HEALTHY (green)	Last ping arrived on time and reported success.
DOWN (red)	Ping is late, or the job reported a failure. An incident is open.
Last ping	When the job last reported; "(reported a failure)" if it called `/fail`.
Run tape / Recent runs	Sentra's evaluation after each ping, and at each deadline — green = on time, red = late or failed.
Incidents	One row per outage: when it opened, how long it lasted, the reason.

Step 7. Demo: the job runs but fails

```
New-Item -ItemType File "$env:TEMP\sentra-demo-backups\FAIL"
schtasks /Run /TN "SentraDemoJob2"
```

MEANING

The `FAIL` file makes the script fail on purpose (like a full disk or database error). The script then calls `/ping/<token>/fail`.

Expected: the log shows `FAIL simulated failure ...`. Within seconds the check turns **DOWN** and an **Open** incident appears: "Job reported a failure".

Fix it:

```
Remove-Item "$env:TEMP\sentra-demo-backups\FAIL"
schtasks /Run /TN "SentraDemoJob2"
```

Expected: a new `OK` log line; within ~5 s the check is **HEALTHY** and the incident shows **Resolved** with its duration.

Step 8. Demo: the job stops running completely

```
schtasks /Change /TN "SentraDemoJob2" /DISABLE
```

MEANING

Simulates a crashed server, a deleted crontab, or a job that silently stopped. No error is produced anywhere — the only signal is that the pings stop.

Expected: "Last ping" stops updating. About **2 minutes** after the last ping (1 min period + 1 min allowed delay), at the exact deadline, the check turns **DOWN**: "No ping for ..., expected every 1 min".

"You cannot detect the absence of an event by watching for an event. Sentra expects the ping, so silence itself becomes the alarm."

Fix it:

```
schtasks /Change /TN "SentraDemoJob2" /ENABLE
schtasks /Run /TN "SentraDemoJob2"
```

Expected: within ~5 s the check is **HEALTHY** again and the incident is **Resolved**.

Useful commands during the demo

```
# What the job did recently
Get-Content "$env:TEMP\sentra-demo-backups\job.log" -Tail 5

# Task status: last run time, last result, next run time
schtasks /Query /TN "SentraDemoJob2" /V /FO LIST

# The backup zips the job made
Get-ChildItem "$env:TEMP\sentra-demo-backups"
```

Part D — Delete the job

Step 9. Remove the scheduled task and the check

```
schtasks /Delete /TN "SentraDemoJob2" /F
```

Expected: `SUCCESS: The scheduled task "SentraDemoJob2" was successfully deleted.`

Then open the check in the dashboard and scroll to **Manage check**:

- ⏸ **Pause** stops monitoring and keeps the history (use ▶ **Resume** to restart it), or
- **Delete check** → **Yes, delete** removes the check with all its runs and incidents.

Expected: after deleting, you land back on the dashboard and the check is gone. Without this step the check would go **DOWN** after ~2 minutes, because the pings have stopped (which is the correct behaviour).

Optional — remove the demo backups and log:

```
Remove-Item "$env:TEMP\sentra-demo-backups" -Recurse -Force
```

Part E — Troubleshooting

Problem	Cause and fix
`mvn` not recognised	Terminal opened before Maven was installed. Run the PATH line in A2, or restart the terminal/IDE.
Backend: "Port 8081 was already in use"	A backend is already running. Use it, or close the other window first.
Backend can't connect to database	Docker Desktop not running or container stopped. Redo A1.
`curl` gives 404	Wrong or incomplete token. Copy it again from the check page (Step 2).
`curl` gives 403 / "Whitelabel Error Page"	The URL is missing `/ping/<token>`. Only that path is public.
Check is DOWN but the task "should" be running	Laptop on battery and Step 5 skipped, or the task is disabled. Check with `schtasks /Query ...`; "Last Run Time 1999" means it never ran.
Two demo jobs fail together	The `FAIL` file is shared by every job that runs `demo-cron-job.ps1`. Delete it to recover all of them.
Task runs but no ping arrives	If the token starts with `-`, PowerShell may read it as an option. Create a new check to get a different token.